1.5.4 Identity property/law for the collection $\mathbb{Q}$ of rational numbers

i) Identity property for Addition

For any rational number a, there exists a unique rational number **0** such that $0 + a = a = 0 + a$.

ii) Identity property for Multiplication

For any rational number a, there exists a unique rational number **1** such that $1 \times a = a = a \times 1$.

Illustration

Take $a = \frac{3}{-7}$ that is, $a = \frac{-3}{7}$. Now $\frac{-3}{7} + 0 = \frac{-3}{7} = 0 + \frac{-3}{7}$ (Isn't it?)

Hence, 0 is the additive identity for $\frac{-3}{7}$

Also, $\frac{-3}{7} \times 1 = \frac{-3}{7} = 1 \times \frac{-3}{7}$ (Isn't it?)

Hence, 1 is the multiplicative identity for $\frac{-3}{7}$

1.5.5 Inverse property/law for the collection $\mathbb{Q}$ of rational numbers

i) Additive Inverse property

For any rational number a, there exists a unique rational number $-a$ such that $a + (-a) = (-a) + a = 0$. Here, 0 is the additive identity.

ii) Multiplicative Inverse property

For any rational number b, there exists a unique rational number $\frac{1}{b}$ such that $b \times \frac{1}{b} = \frac{1}{b} \times b = 1$. Here, 1 is the multiplicative identity.

Illustration

Take $a = \frac{-11}{23}$ Now, $-a = -\left(\frac{-11}{23}\right) = \frac{11}{23}$

So, $a + (-a) = \frac{-11}{23} + \frac{11}{23} = \frac{-11+11}{23} = \frac{0}{23} = 0$

Also, $(-a) + a = \frac{11}{23} + \frac{-11}{23} = \frac{11-11}{23} = \frac{0}{23} = 0$

$\therefore a + (-a) = (-a) + a = 0$ is true.

Also, take $b = \frac{-17}{29}$. Now, $\frac{1}{b} = \frac{29}{-17} = \frac{-29}{17}$

$b \times \frac{1}{b} = \frac{-17}{29} \times \frac{-29}{17} = 1$. Also, $\frac{1}{b} \times b = \frac{-29}{17} \times \frac{-17}{29} = 1$

$\therefore b \times \frac{1}{b} = \frac{1}{b} \times b = 1$ is true.

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